

THE STATE UNIVERSITY OF ZANZIBAR



SCHOOL OF COMPUTING

DEPARTMENT OF COMPUTER SCIENCES AND INFORMATION TECHNOLOGY

COURSE NAME: HUMAN RESOURCE MANAGEMENT AND INFORMATION SYSTEMS

COURSE CODE: INF 3108

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GROUP TOPIC:

- Data and Data Capturing in HRIS.

1. Introduction

Data and data capture in a Human Resource Information System (**HRIS**) involve collecting, recording, validating, and storing employee-related information in a structured manner. The quality of an **HRIS** largely depends on the accuracy, completeness, and timeliness of the captured data.

Data capture is the first stage of the **HR** information lifecycle because all **HR** functions, such as recruitment, payroll, attendance, performance management, and reporting, rely on accurate employee data.

Human Resources data is any information related to employees, job applicants, contractors, departments, and **HR** activities within an organisation. These data may be included: -

- Personal
- Employment-related
- Financial
- Behavioral
- Performance-based

This information helps Human Resources management make informed decisions regarding employees and organisational performance.

2. Categories of Data Captured in HRIS

A. Personal Employee Information

This information identifies an employee with these **Data Fields, including:** -

- Employee ID
- Full Name
- Address
- Phone Number
- National ID Number

The main Purpose of the personal employee information are: -

- Employee identification
- Communication
- Emergency response

Example of employee information

Field	Value
Employee ID	EMP001
Name	Fatma Ali
Gender	Female
Date of Birth	10/05/1999
Phone	+255777425455

B. Recruitment Data

Information collected during the hiring process.

Data Fields

- Applicant Name
- Educational Qualifications
- Work Experience
- Interview Results

- Recruitment Status

Purpose

- Candidate evaluation
- Recruitment tracking
- Talent acquisition analysis

Example of recruitment Data

Candidate	Position	Status
Sauda Ali	Accountant	Shortlisted
Moh'd Saleh	Teacher	Interviewed

C. Employment Data

Details regarding the employment relationship.

➤ **Data Fields**

- Employment Date
- Job Title
- Department
- Work Location

➤ **Employment Types**

- Permanent
- Temporary
- Contract
- Part-Time

➤ **Purposes of Employment Data to HRIS**

- Workforce management
- Organisational planning
- Employee tracking

D. Payroll Data

Payroll data is used to calculate employee compensation.

➤ **Data Fields**

- Basic Salary
- Overtime Pay
- Allowances
- Bank Account Details

➤ **Purpose of Employment Data to HRIS**

- Salary processing
- Tax compliance
- Financial reporting

Example of payroll Data

Item	Amount
Basic Salary	1,500,000 TZS
Housing Allowance	300,000 TZS
Tax Deduction	150,000 TZS

E. Benefits Administration Data

Tracks employee benefits.

➤ **Data Fields**

- Health Insurance
- Pension Contributions
- Housing Benefits
- Transport Allowances

➤ **Purpose of Employment Data to HRIS**

- Benefits management
- Employee welfare
- Compliance requirements

F. Health and Safety Data

Maintains workplace safety records.

➤ **Data Fields**

- Medical Examinations
- Workplace Injuries
- Safety Training Records
- Incident Reports

➤ **Purpose of Employment Data to HRIS**

- Occupational safety compliance
- Risk management

3. Importance of Data Capturing in HRIS

Accurate data capturing enables:

- Effective payroll processing
- Reliable employee records
- Better decision-making
- Workforce planning
- Performance management
- Faster reporting
- Improved organisational efficiency

4. Sources of HR Data

HR data can originate from several sources.

➤ **Internal Sources**

Within the organisation:

- Employee registration forms
- Payroll systems
- Attendance systems
- Performance appraisal systems
- Training records

➤ **External Sources**

Outside the organisation:

- Government agencies

- Educational institutions
- Recruitment agencies
- Insurance providers
- Tax authorities

5. The Data Capturing Methods for HRIS

1) Manual Data Entry

HR staff manually enter data into the system.

Process

- a) Collect documents.
- b) Verify information.
- c) Enter data into HRIS.
- d) Save records.

Advantages

- Easy to implement
- Low initial cost

Disadvantages

- Human errors
- Time-consuming
- Duplicate records

2) Electronic Forms

Employees complete digital forms online.

Examples

- Employee registration forms
- Leave request forms
- Performance evaluation forms

Advantages

- Faster data collection
- Reduced paperwork
- Better accuracy

3) Scanning and OCR Technology

The **OCR** (Optical Character Recognition) converts paper documents into digital text.

Documents Scanned

- Certificates
- Contracts
- National IDs
- Passports

Benefits

- Reduced manual entry
- Faster document processing

4) Biometric Data Capture

Uses biological characteristics.

Methods

- Fingerprint scanners
- Facial recognition
- Iris recognition

Applications

- Attendance tracking in an organization's employee
- Access control for any data access in an organization

Benefits

- High accuracy
- Eliminates buddy punching
- Enhanced security

5) Integration with Other Systems

Automatic data transfer between systems.

Systems Integrated

- Payroll systems
- Accounting systems
- ERP systems
- Recruitment platforms

Benefits

- Reduced data duplication
- Consistent information
- Improved efficiency

6) Barcode and QR Code Technology

Employees use ID cards containing codes.

Uses

- Attendance recording
- Employee identification
- Access control

Benefits

- Fast processing
- Reduced entry errors

6. Data Validation During the Data Capturing

Validation ensures that only correct data enters the system.

Validation Techniques

➤ Mandatory Field Validation

The Required fields cannot be left blank all must be filled during the data capturing.

Example:

- Employee Name
- Employee ID

➤ Data Format Validation

Must check the correct format accordingly.

Examples:

- Email format

- Phone number format

➤ **Range Validation**

Ensures values fall within acceptable limits.

Example:

- Age between 18 and 65 years

➤ **Data Duplication Checking**

Prevents duplicate employee records.

Example:

- Duplication of National ID number

7. Challenges in Data Capturing

Any organisation may face several problems during the Data capturing process:

Challenges	Details
Human Errors	• Typographical mistakes • Missing information
Duplicate Records	• Same employee entered multiple times
Incomplete Data	• Missing documents • Unfilled fields
Data Security Risks	• Unauthorized access • Data breaches
Technical Problems	• System failures • Network interruptions
Resistance to Technology	• Employees reluctant to use digital systems

8. Best Practices for Effective Data Capturing

➤ **Standardised Data Collection Forms**

Use consistent forms throughout the organisation.

➤ **Employee Self-Service (ESS)**

Allow employees to update personal information.

➤ **Regular Data Audits**

Review data periodically for accuracy.

➤ **Strong Security Controls**

- Password protection
- Role-based access control

➤ **Real-Time Data Capture**

Capture information immediately when events occur.

9. Conclusion

Accurate data capture is essential for an HRIS to provide reliable information for managers and decision-makers, making it a critical function in human resource management. Accurate data capture is vital for an HRIS to deliver reliable information to managers and decision-makers. It is one of the most critical functions in human resource management, and any shortcomings in this area can severely compromise decision-making.

10. References

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